🧊 Thermal Energy & States of Matter

Ice Cream in a Bag Experiment

Grades: 3rd & 5th **Duration:** 45-60 minutes **Subject:** Physical Science

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A kitchen science treat: turn cream into ice cream by pulling heat out and shaking up the fun.

🎯 Learning Objectives

3rd Grade (NGSS 2-PS1-4 introduced; builds toward 5-PS1-2)

Note: Thermal energy and states of matter are primarily addressed in NGSS at 2nd grade (2-PS1-4) and 5th grade (5-PS1-2). This lesson introduces these concepts to 3rd graders as enrichment and prepares them for 5th grade standards.

Students will be able to:

- **Identify states of matter**: Recognize that cream changes from liquid to solid
- **Understand heat transfer**: Explain that heat moves from warmer objects to colder objects
- **Observe cause and effect**: Describe what happened and why
- **Use scientific vocabulary**: Correctly use: *solid, liquid, thermal energy, heat transfer, freezing*

5th Grade (NGSS 5-PS1-2, 5-PS3-1)

Students will be able to:

- **Explain state changes at particle level**: Describe how molecules slow down and lock into place
- **Understand conservation of energy**: Explain that energy moved (wasn't created or destroyed)
- **Analyze the role of salt**: Explain how salt lowers ice's melting point
- **Make real-world connections**: Relate experiment to refrigerators and salting roads
- **Collect and interpret data**: Record temperatures and explain observations

👩‍🏫 **Parent/Teacher Notes:** Read the short science section together before starting so students know what to watch for. For younger students, focus on the simple idea that heat leaves the cream and the cream freezes. For older students, add the idea that salt helps the ice mixture get colder and pull heat away faster. Optional enrichment can happen later, but the experiment works well on its own.

🔬 The Science

🤔 How do we freeze ice cream in a bag?

Ice cream freezes when heat leaves the cream. In this experiment, the cream starts warmer than the salty ice around it, so heat moves out of the cream and into the ice mixture. As the cream loses thermal energy, it changes from a liquid to a cold, thick solid.

🔥 Heat always moves

Heat moves from warmer things to colder things. That means the cream does not get "cold added" to it. Instead, heat leaves the cream and moves into the salt-and-ice mixture.

Quick picture: warm cream → colder salty ice bath

🧊 Diagram 1: Which way does the heat go?

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Warm cream mixture

Starts as a liquid with **more thermal energy**.

heat →

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Colder salty ice bath

Pulls heat away so the cream can freeze.

🧂 Why add salt?

Plain ice sits at about 32°F, but salt helps the ice melt at a lower temperature. That makes the ice bath even colder, so it can pull heat out of the cream faster. This is the same reason salt can help melt icy roads in winter.

5th grade connection: the energy is not lost; it moves from the cream to the ice.

🥄 Why do we shake?

Shaking keeps bringing warmer cream to the cold sides of the bag, which helps freezing happen more evenly and more quickly. It also helps keep the texture smoother by stopping large ice crystals from building up.

Science words: thermal energy, heat transfer, freezing, liquid, solid

🔬 Diagram 2: What changes during the experiment?

🥛 **Liquid cream**

Particles can slide past each other, so the mixture can slosh and pour.

🍦 **Frozen ice cream**

After heat leaves, particles slow down and stay much more locked in place.

❄️🧂 **Salt + ice bath**

The extra-cold bath absorbs heat from the cream while the ice starts to melt.

What students should notice: the ice melts, the bag gets very cold, and the cream changes from liquid to ice cream because heat moved out of the cream.

📦 Materials

Per Student/Group

- ☐ 1 cup half-and-half (or substitute: ½ cup whole milk + ½ cup heavy whipping cream)
- ☐ 2 tablespoons sugar
- ☐ ½ teaspoon vanilla extract
- ☐ 1 small ziplock bag (sandwich size)
- ☐ 1 gallon ziplock bag
- ☐ 3-4 cups ice
- ☐ ⅓ cup rock salt (recommended; do not substitute table salt (it dissolves too fast and may cause saltwater to seep into the inner bag))
- ☐ Towel or winter gloves
- ☐ Spoon for eating!

⚠️ Safety Notes

- Salt + ice mixture gets *very* cold, so use a towel or gloves.
- Double-check bag seals to prevent leaks
- Wash hands before handling food ingredients
- Check for dairy allergies before starting

Optional Equipment

- Thermometer: *strongly recommended for 5th grade;* if no thermometer, replace the temperature rows in the data table with descriptive observations ("feels warm," "feels icy cold")
- Timer
- Measuring cups/spoons

🔬 Procedure

Step 1: Make Predictions (5 min)
Ask: "What do you think will happen when we put liquid cream in a bag surrounded by ice and salt? Why?" Have students write or share their predictions.

Step 2: Prepare the Ice Cream Mix (5 min)
Combine half-and-half, sugar, and vanilla in the small ziplock bag. Squeeze out excess air and seal tightly. Double-check the seal!

Step 3: Record Starting Data (3 min)
If using thermometer: record temperature of cream mixture and temperature of ice. Have students note the state of each (liquid cream, solid ice).

Step 4: Build the "Freezer" (5 min)
Put ice in the gallon bag. Add salt and shake to distribute. Place the small bag of cream inside, surrounded by ice. Seal the gallon bag.

Step 5: Shake! (10-15 min)
Shake and massage the bag continuously. Use a towel, as it gets very cold! Take turns if multiple people are present. Check the cream every few minutes to observe changes.
Solo homeschool tip: Wrap the bag in a kitchen towel and roll it back and forth across a table, or pass it between your hands like a hot potato. The parent can take 2-minute turns with the child to make shaking manageable.

Step 6: Observe Results (5 min)
Open carefully. Observe the cream (now ice cream!). If using thermometer, record final temperatures. Discuss what happened.

Step 7: Discuss & Eat! (10 min)
Use discussion questions below while enjoying the ice cream!

❓ Discussion Questions & Answers

3rd Grade Questions

Q: What state was the cream at the start? At the end?

A: Liquid at the start, solid at the end.

Q: Did we add cold or remove heat?

A: We removed heat. "Cold" isn't something we add; we pulled heat out of the cream and into the ice.

Q: Where did the heat go?

A: The heat moved from the cream into the ice and salt mixture. That's why the ice melted!

Q: Why did we need to shake the bag?

A: Shaking helped the heat transfer faster by making sure all the cream touched the cold sides of the bag. It also made the ice cream smooth instead of chunky.

5th Grade Questions

Q: Why does adding salt make ice colder?

A: Salt lowers ice's melting point (from 32°F to around 0°F or lower). The ice "wants" to melt at this new lower temperature, but to melt it needs energy, so it pulls heat from everything around it, including the cream. This makes the mixture much colder than plain ice.

Q: What's happening at the molecular level when liquid freezes?

A: In a liquid, molecules move freely and slide past each other. When heat is removed, molecules slow down and lock into a fixed pattern (a solid). They're still vibrating, just not moving around anymore.

Q: How is this similar to how a refrigerator works?

A: A refrigerator also removes heat from the inside (your food) and releases it outside (the back coils get warm). Both use the principle that heat flows from warmer areas to colder areas. We created a super-cold environment with salt + ice to pull heat out of the cream.

Q: Why did the ice melt during the experiment?

A: The ice absorbed thermal energy (heat) from the cream. When ice absorbs enough energy, it changes state from solid to liquid. The energy went into breaking the bonds holding ice molecules in place.

Q: Is the total amount of energy in the bag more, less, or the same at the end?

A: The same! Energy isn't created or destroyed (conservation of energy). It just moved from the cream to the ice. The cream lost thermal energy, the ice gained it.

📊 Student Worksheet

No thermometer? Replace the temperature rows with descriptions like "warm," "cold," "ice cold," or "frozen." The observation row is for anything you notice: texture, sounds, appearance.

Measurement	Start	After 5 min	After 10 min	End
Cream Temperature				
Cream State (liquid/solid)				
Ice Temperature				
Observations				

📝 Student Notes / Predictions

🚀 Extension Activity

Try the experiment *two* ways: one bag *with* salt, one *without* salt. Compare how long each takes to freeze. The no-salt bag won't freeze properly. Why not?

Expected result: Without salt, ice stays at 32°F, which is not cold enough to freeze ice cream quickly. With salt, the mixture drops well below freezing, pulling heat out of the cream much faster.

✂️ PARENT/TEACHER ANSWER KEY

Detach before giving worksheet to students

Expected Data Table Results

Measurement	Start	After 5 min	After 10 min	End (15 min)
Cream Temperature	~70°F (room temp)	~40-50°F	~32-35°F	~25-30°F
Cream State	Liquid	Liquid (thickening)	Slushy/semi-solid	Solid (ice cream!)
Ice Temperature	~32°F	~10-15°F	~5-10°F	~0-10°F
Observations	Bag is cold	Bag is *very* cold; ice melting.	Cream getting thick	Cream is solid!

Note: Actual temperatures vary based on amount of salt, ice quality, shaking vigor, and ambient temperature. The pattern (decreasing cream temp, frozen end state) should be consistent.

Core Concepts: What Students Should Learn

3rd Grade Level:

- Heat always moves from hot things to cold things
- Freezing happens when heat is removed from a liquid
- Salt makes ice colder than regular ice
- Shaking helps the cream freeze evenly and faster
- The cream changed from a liquid state to a solid state

5th Grade Level (deeper understanding):

- Thermal energy is the motion of particles; more heat = faster particles
- When liquid freezes, molecules slow down and lock into a fixed structure
- Salt lowers the freezing point of water (colligative property)
- Energy is conserved (it moved from cream to ice, but total energy stayed the same)
- Melting and freezing are opposite processes involving energy absorption/release

Discussion Question Guidance

For All Students:

- **Q: What state was the cream at the start vs. end?**
  A: Liquid → Solid. Accept any phrasing that shows understanding of state change.
- **Q: Did we add cold or remove heat?**
  A: Removed heat. Emphasize: "cold" isn't a thing we add; it's the absence of heat!
- **Q: Where did the heat go?**
  A: Into the ice (which melted). Students should connect cream getting colder = ice melting.
- **Q: Why shake the bag?**
  A: Speed up heat transfer, ensure even freezing, prevent large ice crystals (chunky texture).

For 5th Graders (additional depth):

- **Q: Why does salt make ice colder?**
  A: Salt lowers ice's melting point. The ice needs energy to melt at the new lower temp, so it pulls heat from surroundings. This creates a super-cold environment (<0°F).
- **Q: What's happening at molecular level?**
  A: Liquid molecules slide freely. Removing heat slows them until they lock into a rigid structure (solid). Molecules still vibrate in place but don't move around.
- **Q: How is this like a refrigerator?**
  A: Both remove heat from one area and release it elsewhere. Refrigerators use a coolant cycle; we used salt-ice as our "coolant."
- **Q: Why did ice melt?**
  A: Ice absorbed thermal energy from the cream. Absorbed energy breaks molecular bonds, changing ice (solid) to water (liquid).
- **Q: Total energy at end?**
  A: Same as start! Conservation of energy. Energy moved (cream → ice) but wasn't created/destroyed.

Troubleshooting Common Issues

- **Ice cream won't freeze:** Not enough salt, not enough ice, bag too large (small inner bag works better), or didn't shake enough. Temperature of salt-ice mix should drop well below 20°F.
- **Ice cream is grainy/icy:** Shook too little or too slowly. Slow freezing = big crystals. Fast freezing with vigorous shaking = smooth texture.
- **Bag leaked:** Double-bag or use freezer zip bags. Test seal before adding to outer bag.
- **Too salty taste:** Small hole in inner bag. Discard and remake. Rinse outside of inner bag before opening if worried about contamination.

What Success Looks Like

3rd Grade:

- Can identify the cream changed from liquid to solid
- Can explain heat moved from cream to ice
- Can state that salt makes ice colder

- Uses vocabulary: thermal energy, heat transfer, freezing, states of matter

5th Grade:

- Can explain particle behavior in solids vs. liquids
- Can describe how salt lowers freezing point
- Can apply conservation of energy to the system
- Can make connections to real-world applications (refrigerators, road salt, etc.)
- Can use precise scientific language in written explanations

Extension Activity Answer

Bag *without* salt: Will not freeze properly or will take much longer (more than 30 min). Plain ice stays at 32°F, which isn't cold enough to freeze cream quickly. The cream may get cold but remain liquid or become slushy at best.

Bag WITH salt: Freezes in 10-15 minutes. Salt-ice mixture drops to 0-10°F, creating a powerful "heat sink" that rapidly pulls thermal energy from the cream.

Conclusion: Salt is essential for making ice cream! It's not just for flavor; it's the science secret that makes freezing possible without an electric freezer.

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